

Reviewer Status Report

Updated local build, 24 September 2026. Scientific limitations remain distinct from source verification. Portal confirmation of the author-selected baseline, author metadata reconciliation and package approval are pending.

R1-1 single collection, 31 images

Reviewer concern: Single dataset source and an extremely small manually verified benchmark: All experiments are carried out on the CGHD-1152 dataset only. The self-built human-verified benchmark merely contains 31 images with wide confidence intervals and insufficient coverage of sample distributions, which cannot prove that the proposed method can be universally adapted to various hand-drawing styles.

Action taken: 1. Action taken: results labeled as within-collection only. Drafter groups presented descriptively (**16 direct and 15 inferred** assignments disclosed, per-drafter table included): different hands give some style variation, but same corpus, so no universal-adaptation claim and no held-out-drafter experiment. Residual label bias disclosed as mitigated but not eliminated, with independently authored synthetic data as the second leg; future expansion described as future work. **34-annotated to 31-scored** construction disclosed with exclusion reasons; join micro-F1 **0.890** with the 95 percent bootstrap interval stated outright. **2. Why:** N=31 cannot support a universal claim, and pretending otherwise fails verification. Honest bounding plus the synthetic second leg preserves credibility. **3. Alternative rejected:** rushed expansion to 100-plus images, or a held-out-drafter experiment on tiny partly-inferred cells. Would produce noisy labels, risk missing the freeze, and read as tuned to the test set. Low-quality new data lowers odds more than clean disclosure.

Remaining: Owner read-through. RE-VERIFY: 34-annotated to 31-scored accounting plus the 16-direct/15-inferred drafter chain (Bosco).

R1-2 complex parts

Reviewer concern: The algorithm only supports basic passive and active discrete components: Complex components such as transformers, thyristors and optocouplers can only be located, while their pin connection relationships cannot be extracted, limiting the range of component types supported by netlist generation.

Action taken: 1. Action taken: new capability table separating three things the old text blurred together: • tested connectivity • estimated pins • illustrative export plus a metric-boundary row. Benchmark scope (R/C/L/D/Q plus voltage-source and IC groups) stated without validating universal device templates; named transformer type two-pin geometric guess, optocoupler templates, and unknown-type fallback pins disclosed without validated electrical pin semantics. Switch behavior stated exactly as the code does it: shorter OBB-edge midpoint pins with long-axis AABB fallback, fixed **0.001-ohm** emission only if pins 0 and 1 map to distinct nodes, otherwise no switch line, no state or control model; ICs and listed complex devices with no export model skipped. Output and evaluation scope clarified in abstract, pipeline overview, metric definition, and conclusion. **2. Why:** the detector merges original classes, so templates cannot establish separately detected device identities, and there is no ground truth to score complex-device pins. Exact code-truth wording replaces a blanket exclusion with something checkable. **3. Alternative rejected:** implement full pin extraction for transformers, thyristors, and optocouplers in this revision. Unverifiable new capability with no evaluation data increases false-claim risk and would be caught on verification.

Remaining: Owner read-through.

R1-3 hand-set thresholds

Reviewer concern: Reliance on manual tuning of fixed scale-relative thresholds: Hyperparameters including τ_{join} , τ_{t} and α need to be manually configured according to the median diagonal length of circuit components. Without an adaptive parameter learning mechanism, the detection accuracy tends to decline when processing circuits of extreme sizes.

Action taken: 1. Action taken: all values published plainly: • multipliers **0.62/0.30/0.20** times component scale, clamped to 24-60/11-28/8-20 pixels • alpha **0.35** fallback-only directional scoring with formula, used only when no component is assigned • completion reach **4x** the clamped pin scale Existing reach-factor sweep 3 to 5 yielding macro-F1 **0.895-0.903** reported; fixed-pixel base **0.820** versus scale-relative **0.816** kept as observed; method descriptions rewritten from the implementation; extreme sizes and controlled rescaling admitted as untested. **2. Why:** adaptive parameter learning is a new research contribution, not a revision

task, and cannot be validated on N=31 without looking tuned to the test set. Publishing the numbers plus the existing sweep bounds the claim with data already in hand. **3. Alternative rejected:** build an auto-tuner or claim robustness to extremes. A rushed learner risks inconsistency with the frozen 192/192 numeric gate and invites questions it cannot answer.

Remaining: Nothing.

R1-4 no values, cannot simulate

Reviewer concern: Lack of a closed-loop framework for component value recognition: The pipeline only reconstructs topological connections of circuits without integrating OCR to identify parameters of resistors, capacitors and voltage sources, so it cannot directly export complete simulation netlists that can be imported into SPICE.

Action taken: 1. Action taken: scope stated in abstract, intro, and conclusion; retitled to **Structural Circuit Netlists** including running heads and pipeline diagram node text; export labeled illustrative throughout, with external values and device models required and real-scan simulation equivalence unvalidated; related-work LTspice-validation implication on real scans removed; export substitutions and unsupported primitives disclosed. **2. Why:** image-derived values and device models are outside the evaluated task. Keeping a simulatable-SPICE claim while reading no values is a false claim and an editor-level defect. **3.**

Alternative rejected: rush an OCR-plus-values loop to earn the old title. Low-quality OCR adds a new error source with no validation, and expands scope instead of fixing the discrepancy.

Remaining: Nothing.

R1-5 small circuits only

Reviewer concern: Inability to handle complex circuits with multi-layer crossings and dense buses: The test samples adopted in this paper are dominated by simple topologies with 3 to 6 components. No dedicated verification is conducted for circuits with high-density multi-track buses and multi-layer crossed wiring, leading to a significant drop in recall on complex circuits.

Action taken: 1. Action taken: limits stated; true distribution printed in the caption (**3-14** components per image in the evaluated electrical subset, median **7**; 15 images at most five, 12 at least ten); dense buses and multilayer crossings stated as untested with nothing claimed beyond the data; crossover discussion narrowed; histogram asset preserved. **2. Why:** the actual counts refute the 3-6-only premise, but count breadth does not validate buses or multilayer crossings. Descriptive counts plus an explicit boundary close the point honestly. **3. Alternative rejected:** claim dense-bus support, or run an unplanned dense-bus evaluation with no data. No ground truth exists for that structure in the benchmark.

Remaining: Nothing.

R1-6 memristor citations

Reviewer concern: It is suggested that the authors cite two papers in the sections related to SPICE simulation of memristive analog circuits: "A Memristor-Based Neural Network Circuit with Classical Conditioning and Fear Generalization" and "Biologically Plausible Memristive Decision-Making Circuit for Adaptive Control in Industrial Autonomous Navigation". Both papers complete full SPICE netlist modeling and simulation verification for memristive circuits, which complement the EDA technical route of converting hand-drawn schematics to SPICE netlists proposed in this paper in application scenarios, and can enrich relevant literature support for digital parsing and simulation deployment of analog memristive circuits.

Action taken: 1. Action taken: both papers checked at abstract level and cited in Related Work with full references (Gao et al., IEEE Trans. Consumer Electronics, vol. 72, no. 2, pp. 3213-3224, May 2026; Gao et al., IEEE Trans. Industrial Informatics, vol. 22, no. 6, pp. 4801-4812, June 2026) as neighboring SPICE-deployment domains; no comparison or benchmark claim altered. **2. Why:** straightforward compliance at near-zero cost. **3. Alternative rejected:** argue relevance decline to keep the related work tight. Contesting a citation suggestion wastes goodwill for zero gain in a one-shot review.

Remaining: Source citation check complete; final package approval pending.

R2-1 statistics, roadmap

Reviewer concern: The scale of the human-verified benchmark is insufficient, casting doubt on the statistical reliability of the conclusions. The evaluation in this paper covers only 31 images. The authors should explicitly clarify the annotation cost and provide a roadmap for dataset expansion, or alternatively supply additional support through more rigorous statistical testing. If 31 images represent the entirety of currently available data, the wording of the conclusions should be made considerably more conservative.

Action taken: 1. Action taken: sampling in four strata with independent annotation and adjudication described; expansion protocol added as future work (stratified by component count, drafter, crossing/bus structure, and capture conditions, with per-image effort and provenance logging); missing timing honestly noted (single annotator, no per-image timing log, so no cost figure invented and no sample-size commitments); conclusions limited to 31 images from one corpus with bootstrap intervals retained; a CI containing zero stated as nonsignificance, not equivalence. **2. Why:** conservative conclusions on the available data are exactly the fallback the reviewer offered. **3. Alternative rejected:** invent per-image timing or promise a completed expansion. Fabricated cost data and false completion claims are the fastest way to lose editor trust.

Remaining: Owner read-through.

R2-2 VLM comparison

Reviewer concern: There is a fundamental fairness issue in the VLM comparison experiment. The VLM enjoys an "oracle advantage" with respect to critical prior information, whereas the geometric pipeline must bear the burden of detection errors. In the paper, the VLM is provided with ground-truth annotated component bounding boxes; under such conditions, the fact that the VLM still shows no statistically significant difference from the geometric method does not sufficiently support the conclusion that "the geometric method is superior in terms of cost and structural validity."

Action taken: 1. Action taken: relabeled as an oracle-component-box connectivity diagnostic with identical annotated inputs to both methods (geometric method uses detected wires); VLM micro-F1 **0.923** versus **0.890** with exact scores on **21/31** reported; paired difference **+0.033** with 95 percent CI **[-0.009, +0.078]**, inclusion of zero stated as nonsignificance rather than equivalence. Removed from the manuscript: • cost multiples and superiority claims • free-form-output criticism • autonomous bottleneck claims • global structural guarantees • the provisional autonomous paragraph (unauditable provenance) VLM section, intro comparison, abstract, real-join caption, discussion, and conclusion revised. **2. Why:** the reviewer correctly identified the asymmetry. Keeping any superiority or cost claim under oracle conditions guarantees rejection. **3. Alternative rejected:** keep the superiority framing, or run a new end-to-end VLM comparison with unauditable provenance. Both preserve the exact sentence the reviewer flagged.

Remaining: Nothing.

R2-3 title overclaims

Reviewer concern: The paper title claims to generate SPICE netlists, yet the paper explicitly acknowledges that it "makes no attempt to read component values." This implies that the current system outputs a topological netlist rather than a simulatable SPICE netlist, which constitutes a notable discrepancy with the claims made in both the title and the abstract.

Action taken: 1. Action taken: retitled everywhere to **Structural Circuit Netlists**, including both running-head arguments in both manuscript sources and the pipeline diagram node text; metric boundary added across abstract, intro, pipeline overview, metric definition, capability table, and conclusion (component-pair metric does not certify pins, net partition, shorts, or simulation equivalence). **2. Why:** title-body mismatch is the defect editors check first. Fixing it everywhere removes the objection instead of arguing it. **3. Alternative rejected:** keep the SPICE title as conventional shorthand. That concedes the factual premise and would certainly fail.

Remaining: Pipeline PDF rebuilt from corrected TikZ; wire chart retained at 0.845/0.758. Final package approval pending.

R2-4 crossing wires

Reviewer concern: The impact of low recall in a critical component category on netlist correctness has not been sufficiently analyzed. The recall of the crossover class in component detection is only 70.7%, substantially lower than that of other categories. Misclassification of crossovers will introduce systematic errors in electrical connectivity, causing two wire segments that should remain electrically independent to be erroneously merged into the same net.

Action taken: 1. Action taken: full two-edge tradeoff reported (C242 suppressing two crossing-adjacent endpoint-endpoint edges: **27/4/0 to 19/0/8** — removing four false pairs loses eight true pairs, so no mitigation gain claimed); audit covers **13 annotated crossovers** across eight images with the stored artifact kept; single-edge, rarity, and safe-suppression interpretations replaced; detector-miss effect explicitly retained as unmeasured because the join-isolation evaluation uses annotated boxes. **2. Why:** suppression is not a mitigation, and attributing observed false pairs to detector recall would be causal overreach. A conditional joining analysis is what the design can measure. **3. Alternative rejected:** retrain the detector to lift crossover recall, or claim the audit resolves the detector pathway. Retraining risks cross-class regressions before the freeze; the causality is unmeasured by construction.

Remaining: Detector-error effect remains unmeasured; partial scientific response.

R2-5 ablations

Reviewer concern: The ablation study is severely insufficient.

Action taken: 1. Action taken: real table as observed: • base-graph/full-pipeline interventions for T-junctions, rail taps, directional preference, and base tolerance scaling • fixed-pixel base **0.820** versus scale-relative **0.816** and full-pipeline ties at **0.890 (418/37/66)** presented as-is without spin • ties stated as neither identical connections nor proof of dispensability Activation and identical-connection explanations plus the unsupported quantified occlusion benefit removed; method description rewritten (graph selection, mandatory slot/dummy assignment, guarded application); missing full-occlusion and completion-wide scale tests disclosed. **2. Why:** keeping the unfavorable base result and the ties as an explicit negative result buys credibility. Editing hurtful rows breaks the 192/192 numeric gate. **3. Alternative rejected:** drop the fixed-pixel row, or run new full-occlusion and scale-removal ablations that re-engineer the pipeline pre-freeze. Rushed new ablations risk inconsistency across the frozen evidence chain.

Remaining: Full occlusion and completion-wide scale ablations remain untested; partial scientific response.

R2-6 single-scalar limit

Reviewer concern: The setting of $\tau = k \cdot s$ may fail when component sizes within a single schematic vary substantially, and no relevant discussion addressing this limitation is provided.

Action taken: 1. Action taken: limitation stated and retained as asked — one scalar per image does not model intra-image size variation (large IC mixed with small discretized); clamps bound tolerances without adapting to heterogeneous symbols; fixed-pixel ablation changes the base graph only while completion retains clamped scale dependence; no rescaling experiment claimed; method text and sweep/ablation interpretation scoped accordingly, label anchor added. **2. Why:** the reviewer asked for discussion, and discussion is full compliance at zero risk. **3. Alternative rejected:** rebuild tau as per-component adaptive in this revision. That is a new method requiring new validation, with no ground truth designed to score it.

Remaining: Nothing.